📘 SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Internal POS & Inventory Monitoring System

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1. INTRODUCTION

1.1 Purpose

This document specifies the functional and non-functional requirements for an Internal POS & Inventory Monitoring System designed to replace the current Excel-based workflow used for inventory tracking, costing, and sales monitoring.

The system will be used exclusively by internal staff and administrators.

1.2 Scope

The system will:

- Track raw material purchases
- Maintain real-time inventory levels
- Calculate production cost dynamically
- Record sales transactions
- Support partial payments
- Monitor revenue and profit
- Provide operational reports

The system will NOT:

- Provide online ordering
- Be accessible to customers
- Support public-facing features

1.3 Definitions

- **POS** – Point of Sale system for internal sales entry.

- **Costing Method** – Algorithm used to compute material unit cost.
- **Reservation** – Temporary allocation of inventory for production.
- **Production Status** – Stage of product preparation.
- **Payment Status** – Status of payment collection.

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2. OVERALL DESCRIPTION

2.1 Product Perspective

The system replaces manual Excel sheets used for:

- Inventory tracking
- Cost computation
- Sales reporting
- Revenue calculation

The system will use:

- React + Ant Design (Frontend)
- Node.js Backend
- PostgreSQL Database

2.2 User Classes

Admin

- Full control of system
- Can configure costing strategy
- Access financial reports

Staff

- Record sales
- Update production status
- Record payments
- View inventory

2.3 Operating Environment

- Web-based internal application
- Desktop-first
- Modern browsers (Chrome, Edge)
- Hosted on internal server or cloud server

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3. SYSTEM FEATURES & FUNCTIONAL REQUIREMENTS

3.1 Inventory Management

3.1.1 Raw Material Management

The system shall allow Admin to:

- Add new raw materials.
- Edit raw materials.
- Define unit type.
- Set low stock threshold.

The system shall:

- Maintain stock_quantity.
- Recalculate unit_cost automatically.

3.1.2 Purchase Logging

The system shall allow:

- Recording of material purchases.
- Entry of quantity and total cost.

The system shall:

- Increase stock_quantity.
- Update unit_cost based on selected costing method.

3.2 Costing Strategy Module

The system shall allow Admin to select one of the following costing methods:

1. Weighted Average Cost
2. Latest Purchase Cost
3. Manual Override

The system shall:

- Apply selected costing method to future transactions.
- Preserve historical sale cost snapshots.

3.3 Product Costing Module

The system shall allow:

- Creation of product templates.
- Assignment of materials per product.
- Specification of material quantities.
- Configuration of labor cost (fixed or percentage).

The system shall automatically compute:

Production Cost =
Σ (material_unit_cost × quantity) + labor

3.4 POS / Sales Module

The system shall allow staff to:

- Create new sale entries.
- Select product.
- Set quantity.
- Override selling price (optional).
- Record notes.

The system shall:

- Capture production cost snapshot.
- Calculate total amount.

- Track production status.
- Track payment status.

3.5 Production Status Workflow

The system shall support the following statuses:

- Confirmed
- In Production
- Ready for Pickup
- Completed
- Cancelled

When status changes to "In Production":

- Required materials shall be reserved.

When status changes to "Completed":

- Reserved materials shall be deducted.

When status changes to "Cancelled":

- Reserved materials shall be released.

3.6 Payment Module

The system shall allow:

- Recording multiple payment transactions per sale.
- Tracking payment method.
- Partial payments.

The system shall:

- Calculate total_paid.
- Calculate remaining_balance.
- Update payment_status automatically.

3.7 Dashboard Module

The system shall display:

- Daily sales revenue
- Monthly revenue
- Monthly profit
- Pending sales
- Low stock materials

The system shall provide visual charts for:

- Revenue trend
- Profit trend
- Top-selling products

3.8 Reporting Module

The system shall generate:

- Daily Sales Report
- Monthly Revenue Report
- Profit Report
- Inventory Valuation Report

Reports shall be exportable to:

- CSV
- Excel

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4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interface

Frontend shall use Ant Design components:

- Table
- Form
- Modal
- Drawer

- Layout
- Statistic
- Badge
- Tabs

UI shall be:

- Clean
- Data-focused
- Optimized for fast entry

4.2 Database Interface

System shall use PostgreSQL with relational structure.

Foreign keys shall enforce data integrity.

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5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

- UI response time < 500ms
- Cost recalculation < 1 second
- Support at least 5 concurrent users

5.2 Security

- Role-based access control
- JWT authentication
- Password hashing
- Secure API endpoints

5.3 Reliability

- Automatic database backups
- Data consistency enforcement

- Transaction-based stock updates

5.4 Usability

- Minimal training required
- Similar workflow to existing Excel
- Clear status indicators

5.5 Maintainability

- Modular backend structure
- Separated services for inventory, sales, payments
- Configurable costing logic

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6. DATA REQUIREMENTS

Main tables:

- users
- raw_materials
- purchases
- products
- product_materials
- sales
- sale_material_snapshot
- transactions
- inventory_reservations
- system_settings

Each sale must store:

- production_cost_snapshot
- selling_price_snapshot

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7. FUTURE ENHANCEMENTS

- Multi-branch support
- Barcode scanning
- Supplier management
- Accounting integration
- Tablet POS mode

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8. APPROVAL

This SRS defines the baseline requirements for development and testing.